SYLLABUS-ALIGNED PROBLEM SET

Problem Set 01

Matrix Arithmetic and Determinants

Show your calculations and reasoning. Write every solution by hand on paper. Upload a clear photograph of each handwritten solution to its matching field in Microsoft Forms. Check your work before submitting.

Problem 1

Let A be 2×3, B be 3×2, and C be 2×2. State which of $AB, BA, A + C, AB + C$ are defined and give their dimensions.

Problem 2

Calculate $2A - 3B$ for the matrices

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ -2 & 0 \end{pmatrix}.$$

Problem 3

For the matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix},$$

find AB and BA. Does multiplication commute for this pair?

Problem 4

Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 2 \end{pmatrix}.$$

Calculate $(AB)^T$ and verify $(AB)^T = B^T A^T$.

Problem 5

Classify the matrices

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}.$$

Identify diagonal, scalar, symmetric and skew-symmetric matrices; more than one label may apply.

Also classify

$$T = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad I_2, \qquad 0_{2\times 3}$$

as triangular, identity or zero matrices, stating their dimensions.

Problem 6

For which real values of t is the matrix

$$A(t) = \begin{pmatrix} t & 2 \\ 3 & t-1 \end{pmatrix}$$

singular?

Problem 7

Calculate

$$\det \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix}$$

using Sarrus' rule. State why this rule is applicable.

Problem 8

For the matrix

$$A = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix},$$

find the minor M_{23} and cofactor C_{23}. Explain their sign difference.

Problem 9

Calculate

$$\det \begin{pmatrix} 0 & 2 & 0 \\ 1 & -1 & 3 \\ 4 & 0 & 2 \end{pmatrix}$$

by Laplace expansion along the first row. Check the result by expanding down the first column.

Problem 10

Let A, B be 3×3 matrices with $\det A = -2$ and $\det B = 5$. Find $\det(AB)$ and $\det(2A^{T})$. A matrix C is formed from A by swapping two rows and then multiplying one row by 3; find $\det C$.